

Mason Drumwright

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EDUCATION

University of Colorado Boulder – College of Engineering

Anticipated: May 2027

B.S. in Aerospace Engineering

Minor in Engineering Management

- **Relevant Courses:** Aerospace Software, Statistical Estimation for Dynamic Systems, Aircraft Dynamics, Orbital Mechanics and Attitude Dynamics, Electronics and Communications, Thermodynamics, Aerodynamics

EXPERIENCE

Research Assistant

Boulder, CO

Autonomous Vehicle Systems Lab

05/26 - Present

- Extending spacecraft dynamics to support the branching dynamics of rigid body subcomponents relative to the spacecraft hub by deriving the general translational and rotational equations of motion for complex spinning and translating systems
- Implementing a new Basilisk Module that simulates 1-DOF point mass motion attached to a prescribed component that is attached to the spacecraft hub by deriving the dynamics and adding them to the software
- Working on simulating solar panel configuration deployments using the prescribed motion state effector module in Basilisk to analyze the results of different panel geometries in Python
- Using a shared Git repository to keep work organized and keep track of edits through version control so that the codebase is safe and new implementations can be easily added

Autonomous UAS and Rover - Dynamics and Controls Lead

Boulder, CO

Senior Project

08/26 - Present

- Leading the Dynamics and Controls team by conducting trade studies, delegating tasks to team members, and working with the hardware team to efficiently develop a robust control system for an autonomous path-following rover.
- Simulating the dynamics for the rover design, including the X-Y planar movement and the payload delivery mechanism, by defining design requirements and deriving equations of motion to visualize motion using MATLAB and Simulink.
- Doing analysis on LQR/LQT and PID control systems by performing Monte Carlo simulations on rover dynamics with varying geometric paths and comparing cross-track error, heading error, and final position error.

RELEVANT PROJECTS

Fixed Wing UAV Build

Boulder, CO

Personal Project

4/26 – 09/26

- Programmed a custom flight controller on a Teensy 4.1 in C++ that performs PID control based on a 15-state Extended Kalman Filter that fuses a 9-DOF IMU, GPS, and barometer to perform autonomous flight
- Developed guidance algorithms that use cascading PID loops by reading the EKF state to perform autonomous maneuvers including climb, cruise, and loiter.
- Simulated custom fixed-wing UAV geometries in MATLAB by implementing stability derivatives to validate a design by analyzing the aircraft's open and closed-loop responses to perturbations

CU in Space – Competitive Rocketry Club

Boulder, CO

Guidance, Navigation, and Control Team

09/25 - Present

- Contributing to the system-level design of a hybrid rocket targeting a precise apogee of 10,000 ft using active airbrake control to increase drag and reduce velocity
- Designing state estimation algorithms to track the rocket during ascent. Using attitude estimation based on accelerometer and gyroscope sensor fusion with an Extended Kalman Filter to estimate the full state
- Developing a Model Predictive control system using MATLAB and C++ that extends airbrakes based on states from EKF by performing Monte Carlo analysis to understand how the rocket behaves with different control parameters/gains.

University of Colorado Boulder

Boulder, CO

Vehicle Design Lab

01/25 – 05/25

- Team project focused on the design and development of an aerospace vehicle capable of withstanding boost-ascent forces, executing a controlled pitch-over maneuver, and achieving stable gliding flight
- Performed analysis on equations of motion using numerical integration techniques (MATLAB) to complete trade studies, including trade-offs between increased wingspan and weight, lift, and drag for design improvement
- Programmed flight simulations in MATLAB based on different design characteristics in order to predict system performance and optimize the lift-to-drag ratio

SKILLS/ADDITIONAL INFORMATION

- Technical Skills: MATLAB, Python, C/C++, Simulink, Control Systems, Linux, Kalman Filters, GNC, Monte Carlo
- Interests: Autonomous Systems, Navigation and Control Algorithms